

DAT535 – Data Engineering Fall 2026

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# Data Engineering Organization Foundations and Evolution



University  
of Stavanger

# Overview

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- Introductions and Course Plan
- History and Trends
- Major Applications

# People

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## ○ Lecturer: Tomasz Wiktorski

- Aker BP, Adjunct Professor at UiS
  - Principal Engineer Future Data Operations
  - At UiS since 2008
  - Program Director CS/DS – 2018-2022
- Working on: Data Engineering, AI, Data Science, Time Series
- Office: KE E426; Office hours: by request

## ○ Lab engineer: Jayachander Surbiryala

- Head Engineer
- Responsible for i.a. lab/project, data center
- Office: KE D424

# People

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- **<https://wayground.com/join?gc=12642597>**

# Approach and Assessment

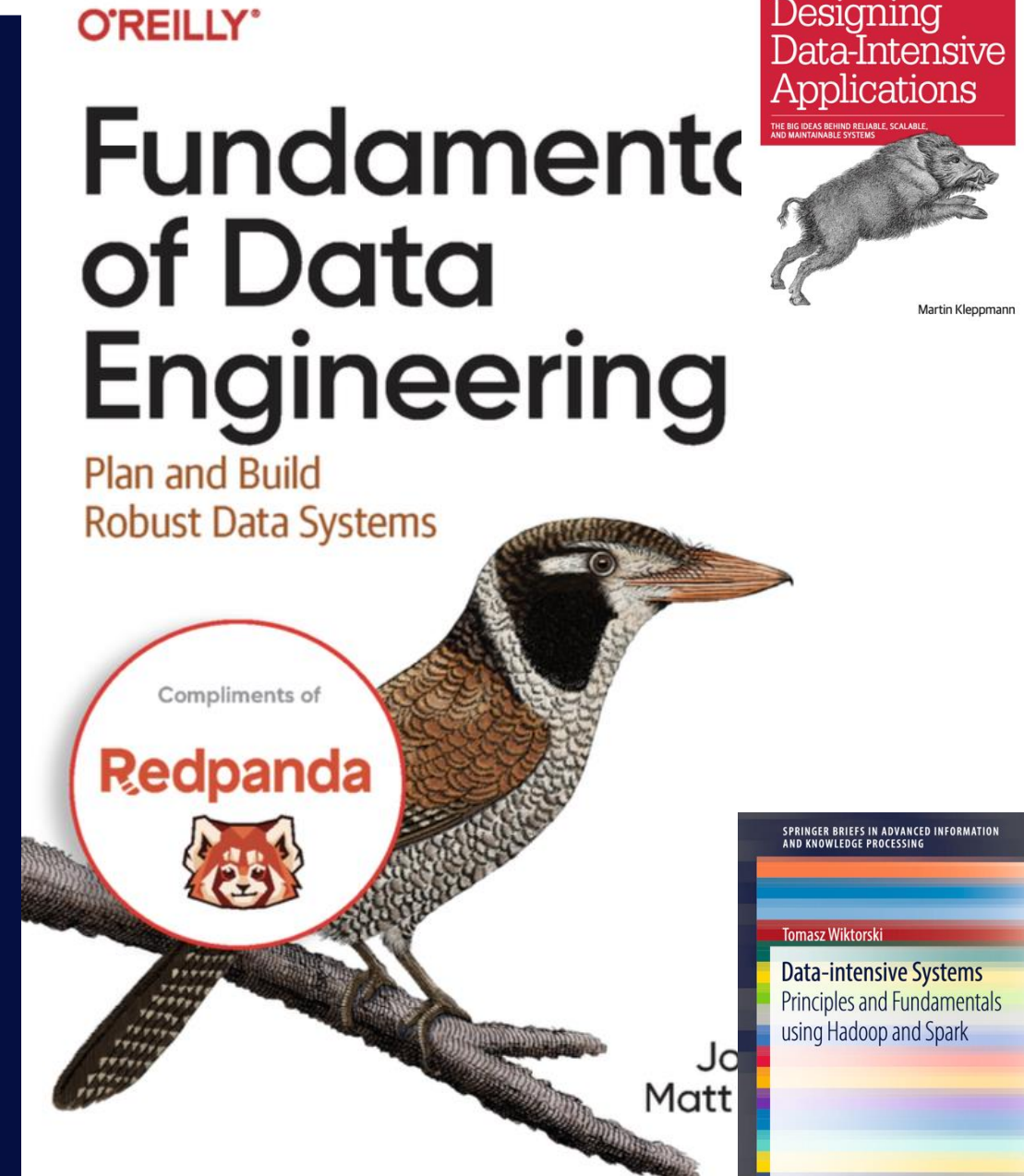
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- What we will and will NOT cover
- How I lecture and work with you...
  - Lectures and guest lectures
  - Group work
  - Labs
  - (Guided) Own study
- My expectations to you...
- Project - Graded
  - Project is completed in groups.
  - If a student fails the project, she/he has to take this next time the course is given.
- Coursework requirements
  - 3 Mandatory Assignments
  - Oral presentations

# Course materials

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- Main book: Fundamentals of Data Engineering
- *Other books:*
  - *Data-intensive Systems – Principles and Fundamentals using Hadoop and Spark*
  - *Designing Data-Intensive Applications*
  - *Learning Spark – Lightning fast data analysis. 2nd Ed*
- Slides
- Materials on Canvas



# High-level Course Plan and Schedule (subject to adjustments)

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- Module 1: Foundations and Architecture
  - Data Engineering Foundations and Evolution
  - Medallion Architecture and Data Engineering Lifecycle
  - Industrial Data Engineering (guest lectures 18/9 and 25/9, Lærdal, Cognite, Reduzer)
- Module 2: Data Processing with Apache Spark
  - Data Ingestion & Bronze Layer
  - Data Cleaning & Silver Layer
  - Data Aggregation & Gold Layer
- Module 3: Automation, Operations and Governance
  - Governance, Security and Compliance (guest 2/10 lecture Equinor)
  - Automation, Monitoring and Reliability (guest 16/10 lecture Zaptec)
- Weekly plan:
  - Lectures and Group Work primarily on Fridays, labs primarily on Mondays
  - Some adjustments will be made to according to course progress and scheduling needs
  - Lectures focus in September, more group work focus in October
  - Week 38 (14-18/9) we will have some adjustments due to conference participation
- 18/9 – 16/10 a series of guest lectures from: Equinor, Zaptec, Lærdal Medical, Cognite, Reduzer
- No lectures during Fall break 5-9/10, extra labs to accelerate work on projects.
- Project Presentations
  - End of October
  - Scheduling dependent on number of groups

# Project Summary

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**Goal:** Build a Medallion Architecture pipeline using Spark (Bronze → Silver → Gold) and deployed with GitHub Actions.

**Scope:** Batch processing only, no streaming or orchestration tools in implementation are obligatory.

**Deliverables:** Spark scripts, report, and a short presentation.

**Check Canvas Assignment for details!**



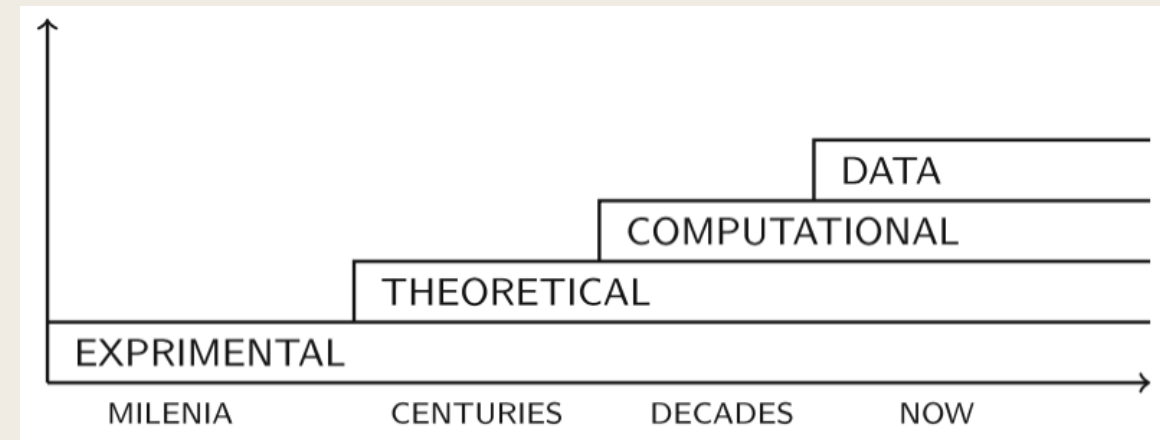
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# History and Trends

# Data as the 4<sup>th</sup> paradigm of science

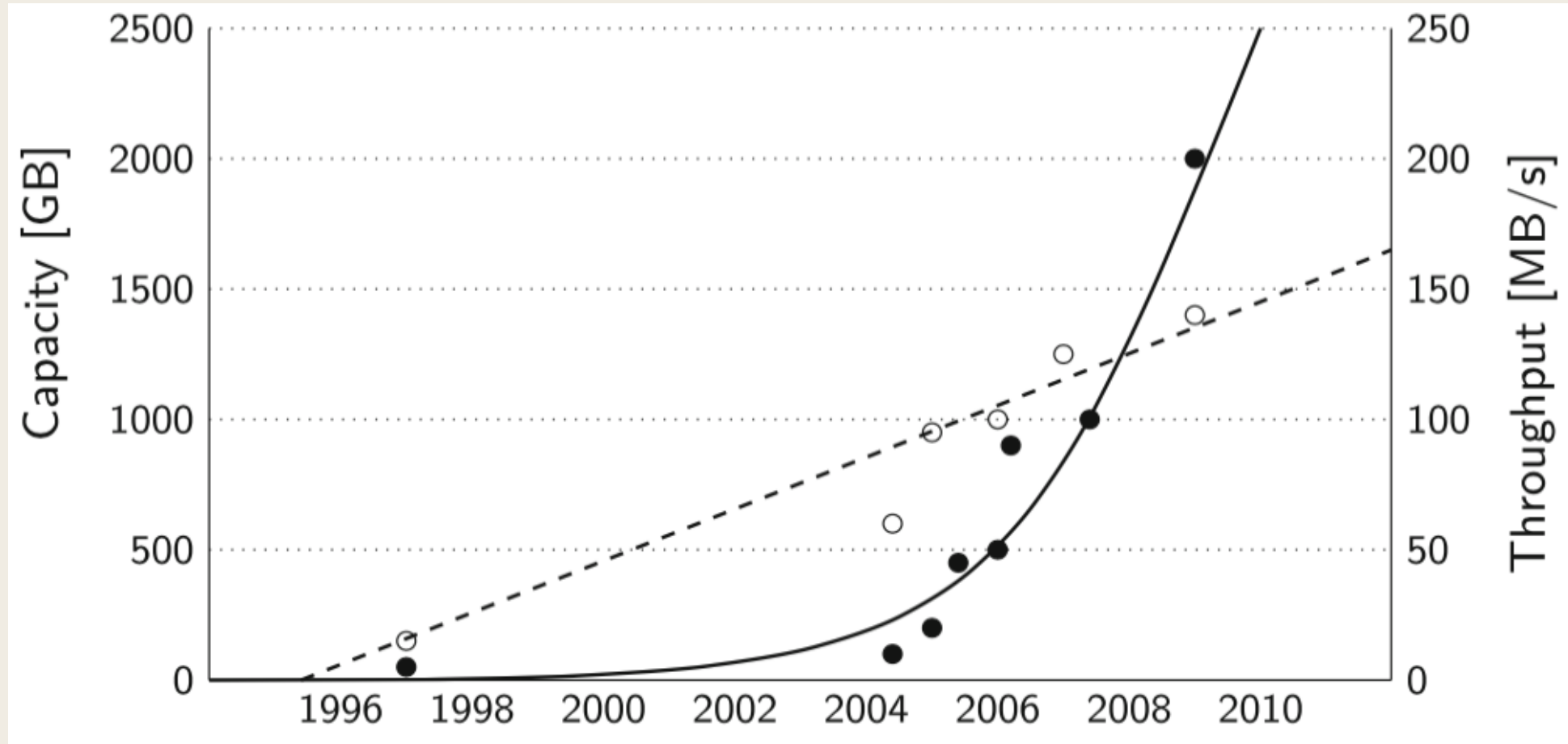
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- 1<sup>st</sup> paradigm – Experimentation
    - Observe and describe
  - 2<sup>nd</sup> paradigm – Theory
    - Develop laws and theories
  - 3<sup>rd</sup> paradigm – Computation
    - Simulate phenomena
  - 4<sup>th</sup> paradigm – Data
    - Better and more sensors
    - Collection capability
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- Paradigms coexist and support one another



*The Fourth Paradigm. Data-Intensive Scientific Discovery*  
(Hey, Tansley, Gray 2009)

# Hardware trends

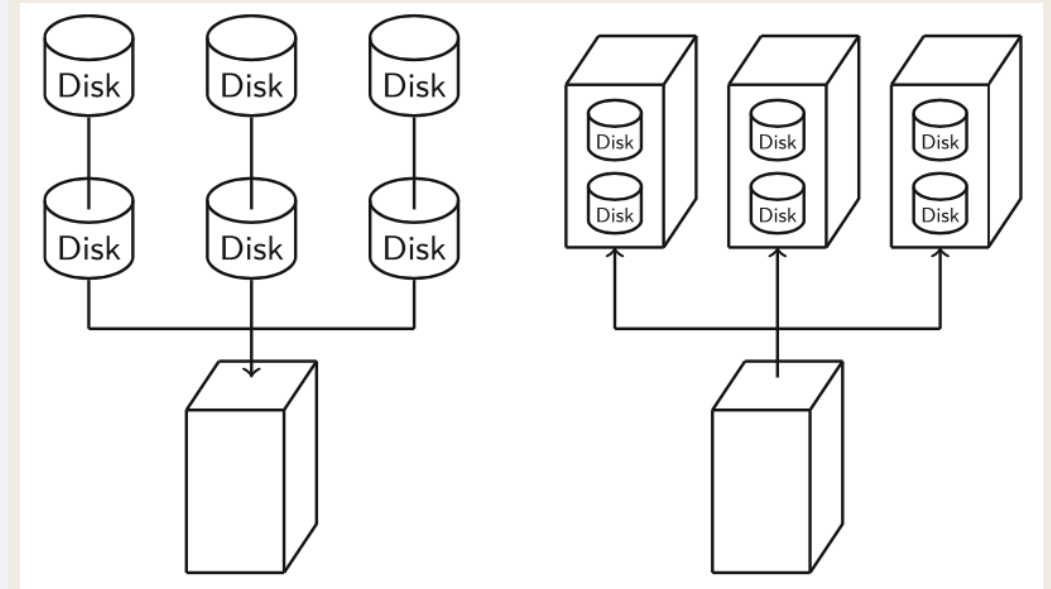
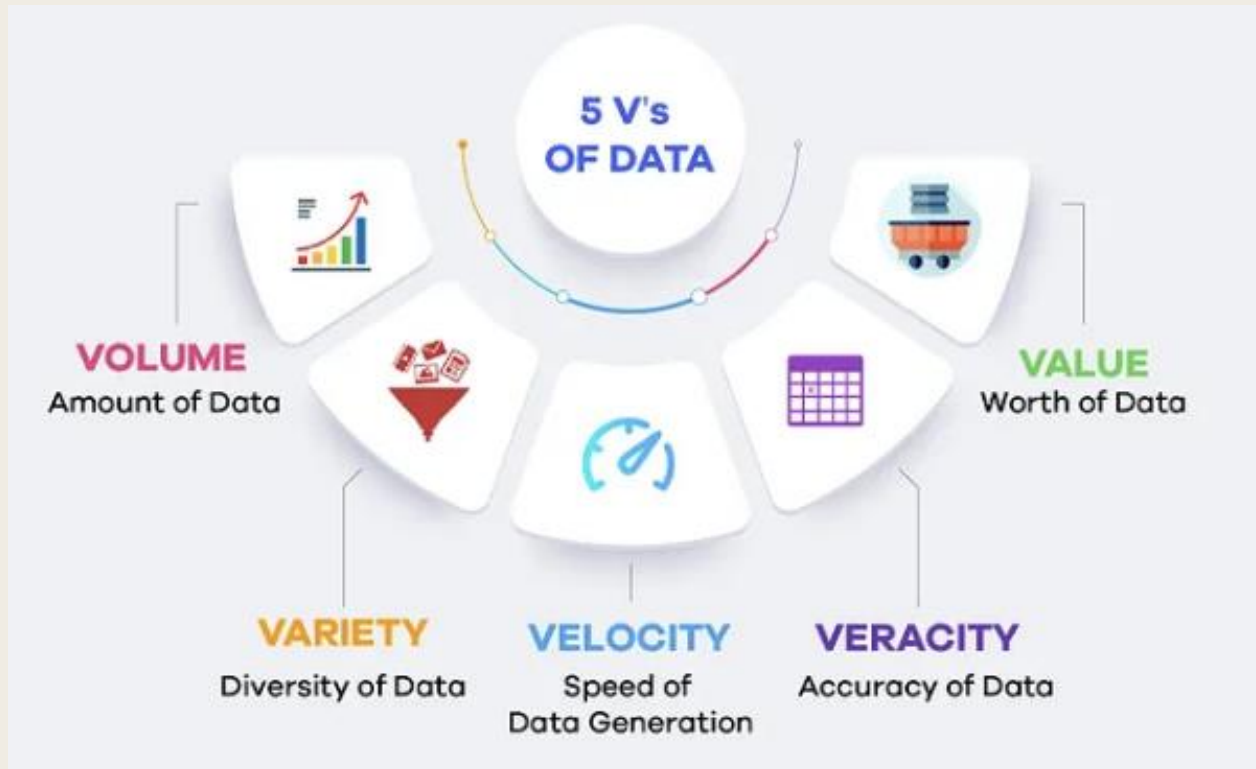


# Data Systems Quick Timeline in Modern Times

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# The V's of Big Data | Scaling Infra and Apps



# Data Engineering Lifecycle and Role

Provides a **holistic framework** for data engineering beyond tools.

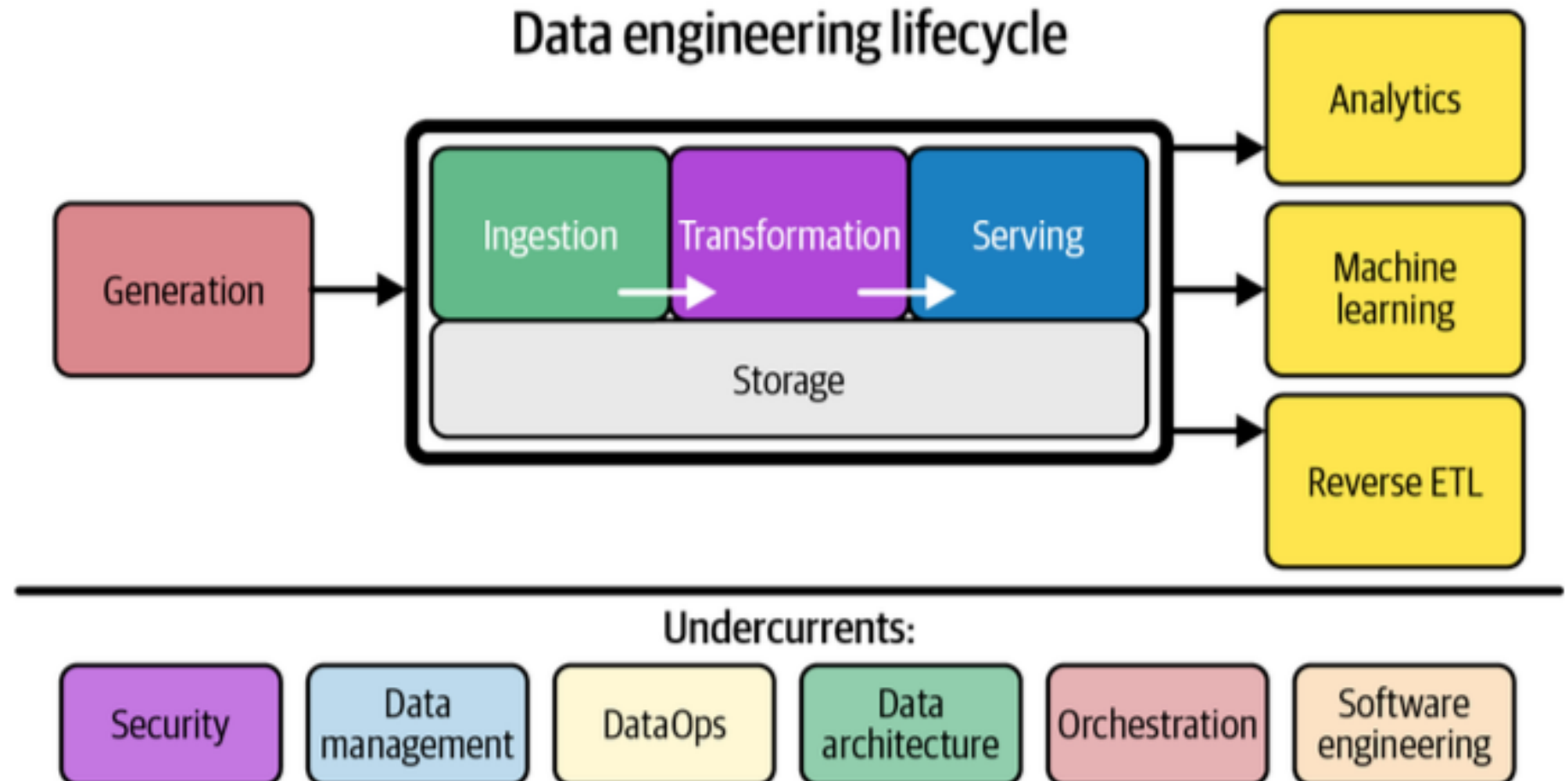
Focuses on **data flow and value creation**, not just technology stacks.

Helps engineers **think in terms of principles**, not just tools.

Ensures **end-to-end accountability** for data quality and reliability.

Bridges the gap between **data engineering and business value**

Storage plays a central and **increasing** role.



# NOSQL

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- No SQL
- Not Only SQL
- Should be: Not Only Relational Data Model
- Literal interpretation can be misleading
- Represents important changes in approach to data modeling

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# Major applications



# Search and Recommendations

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- Major original application
- Processing search index was original motivation for MapReduce
- People you might know – on LinkedIn
- Products you might like – on Amazon



Amazon collection point, White Rose Centre, Leeds, West Yorkshire Mtaylor848 / CC-BY-SA

# Retail Optimization

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- POS, loyalty card, credit card transactions
- Target pregnancy case
- Many of the largest retailers have specialized data labs:
  - Walmart Labs
  - Nordstrom Innovation Lab

MAGAZINE | How Companies Learn Your Secrets



February 16, 2012 - In a preview of this Sunday's New York Times Magazine, Charles Duhigg details how some retailers profit by predicting major changes in your life. By Kassie Bracken on February 17, 2012. [Watch in Times Video »](#)

NYT screenshot under fair use

# Healthcare

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- Reliance on data often not visible at first glance
- Pharmaceutical research
- DNA analysis
- Modeling of neural signals
- Self-monitoring devices like activity bracelets
- >>> Personalized Medicine

"People handling Polar Vantage Multisport GPS Watches" by marcovcrch is licensed with CC BY 2.0.



# Internet of Things

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- Increasing amount and data resolution of sensors and other connected devices
- Increased demand for data storage and computation capabilities to extract value from the produced data
- Energy efficient buildings
- Environmental monitoring
- Agriculture

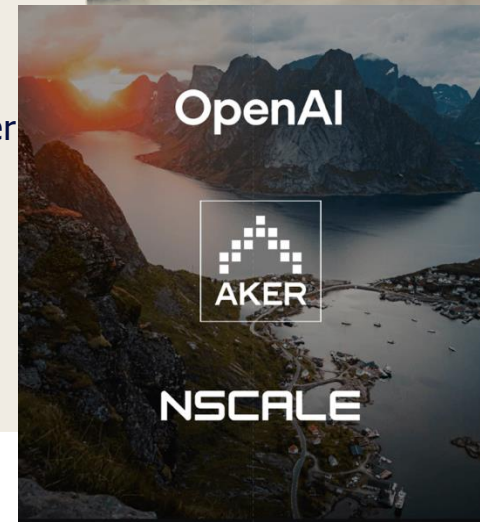




# Data hubs – Data Operations

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- Most common and general use case
- Enterprise and scientific data
- Storage, cleaning, transformation, and archiving – extended ETL
- Analysis can come in addition
- Facebook Hadoop cluster – 1400 nodes, 15PB storage
- CMS Tier-2 at Purdue – 4PB storage
- UiS Research Datacenter (mid 2016) – 250 nodes, 9PB
- Stargate Norway:
  - 230MW of initial capacity powered by renewable hydropower
  - scale by an additional 290MW
  - 100,000 NVIDIA GPUs
- **Major revival with GenAI**
- **Data Engineering's importance growing with GenAI**



# Quiz

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- **<https://wayground.com/join?gc=08710437>**